

ABC BANK

Financial Risk Classification System

FUNCTIONAL REQUIREMENTS DOCUMENT (FRD)

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1. Document Control

1.1 Version History

Version	Date	Author	Description of Change
0.1		Business Analysis Team	Initial draft derived from approved BRD v1.0
0.2		Business Analysis Team	Incorporated feedback from IT and Risk Analyst walkthrough
1.0		Business Analysis Team	Baseline version circulated for stakeholder review

1.2 Approvals

The following stakeholders are required to review and approve this Functional Requirements Document before design and development activities proceed.

Name / Role	Department	Approval Status	Signature & Date
Head of Credit & Lending (Business Owner)	Credit / Lending	Pending	
Chief Risk Officer / Risk Manager	Risk Management	Pending	
Solution Architect	Information Technology	Pending	
QA Lead	Quality Assurance	Pending	
Project Manager	PMO	Pending	
Lead Business Analyst	Business Analysis	Pending	

2. Introduction

2.1 Purpose

This Functional Requirements Document (FRD) defines the detailed functional behavior of the Financial Risk Classification System, including user interactions, system processes, validations, business rules, inputs, outputs, and reporting requirements. It elaborates on the business requirements approved in the Business Requirements Document (BRD v1.0) and serves as the primary reference for designers, developers, testers, and business stakeholders to ensure the solution is implemented and tested in line with approved business requirements.

2.2 Scope

This document covers the functional design of the Financial Risk Classification System, including: applicant data capture and validation, risk factor processing using predefined business rules, risk classification and recommendation generation, loan officer decision workflow, applicant record management, notifications, reporting and dashboards, and system administration (user and rule management). Non-functional characteristics are documented at a level sufficient to guide design; detailed technical architecture is out of scope and will be covered in the Solution Design Document.

2.3 Reference Documents

Document	Version	Purpose
Business Requirements Document (BRD)	1.0	Defines the business need, scope, and high-level business requirements
Existing Lending Policy Manual	Current	Reference for existing eligibility and lending criteria
Data Privacy and Information Security Policy	Current	Reference for data handling and access control requirements

2.4 Document Conventions

Requirements in this document are identified using the following ID conventions to support traceability into design, development, and test artifacts:

Prefix	Category
FR-DC-xxx	Data Capture Requirements
FR-VAL-xxx	Field Validation Requirements
FR-RCE-xxx	Risk Classification Engine Requirements
FR-DEC-xxx	Decision & Workflow Requirements
FR-REC-xxx	Applicant Record Management Requirements

Prefix	Category
FR-RPT-xxx	Reporting & Dashboard Requirements
FR-ADM-xxx	Administration & Configuration Requirements
FR-NOT-xxx	Notification Requirements
FR-AUD-xxx	Audit Trail Requirements
BR-xxx	Business Rule
PF-xxx	Process Flow
SC-xxx	Screen Specification
ERR-xxx	Error Message
NFR-xxx	Non-Functional Requirement
AC-xxx	Acceptance Criterion

3. System Overview

The Financial Risk Classification System provides an automated solution for evaluating loan applicants based on their financial and demographic information. The system collects customer data, validates required fields, processes financial risk factors using predefined business rules, and classifies applicants into risk categories. It presents loan officers with a clear risk assessment, the factors influencing the classification, and a system-generated recommendation (Approve, Manual Review, or Reject). The system maintains applicant records, generates reports, and provides dashboards to monitor loan application trends and risk distribution.

3.1 System Modules

- Applicant Data Capture Module - collects and consolidates applicant financial and demographic data.
- Data Validation Module - validates mandatory fields, formats, and data ranges before processing.
- Risk Classification Engine - applies predefined business rules and historical data to compute a risk score and category.
- Decision & Workflow Module - generates a recommendation and routes applications for approval, manual review, or rejection.
- Applicant Records Management Module - stores and retrieves applicant and application history.
- Reporting & Dashboard Module - provides management reports and visual dashboards of risk and application trends.
- Administration Module - manages users, roles, and risk classification business rules/thresholds.
- Notification Module - sends system alerts and notifications to relevant users.

- Audit Trail Module - logs classification decisions, rule changes, and user actions for compliance.

3.2 User Roles Summary

Role	Description	Primary System Interactions
Loan Officer	Front-line staff initiating and processing loan applications	Enter applicant data, view risk classification, record decisions, view own application history
Credit Committee Member	Reviews higher-value / higher-risk applications	View classification detail, application data, and audit history; record committee decision
Risk Analyst	Maintains risk classification logic	Propose changes to business rules and thresholds; view rule performance reports
Risk Administrator	Approves and publishes rule changes	Approve/reject rule change requests; publish rules; manage risk categories
Compliance / Audit User	Monitors process for regulatory purposes	Read-only access to classification decisions, rule versions, and audit logs
System Administrator	Manages technical configuration and access	Manage user accounts, roles, and system configuration
Branch Manager	Oversees branch lending activity	View branch-level dashboards and reports

4. Process Flows

The process flows below describe the step-by-step interaction between actors and the system for the core scenarios supported by the Financial Risk Classification System.

PF-01: Loan Application Risk Evaluation (Primary Flow)

Step	Actor	Action / System Response
1	Loan Officer	Initiates a new loan application and enters applicant financial and demographic data via the Applicant Data Entry screen (SC-02).
2	System	Validates all mandatory fields and data formats in real time as data is entered (see Section 6, Field Validations).
3	System	On submission, consolidates applicant data with available historical customer data from internal records.
4	System	Applies predefined business rules (Section 5) to calculate a risk score and assign a risk category (Low / Medium / High).
5	System	Generates a recommendation (Approve, Manual Review, or Reject) based on the risk category and applicable thresholds (BR-06).
6	System	Displays the Risk Assessment Result screen (SC-03) to the Loan Officer, showing the risk category, contributing factors, and recommendation.

Step	Actor	Action / System Response
7	Loan Officer	Reviews the result. If the recommendation is 'Approve' or 'Reject' and within the officer's authority, records the decision directly.
8	System	If the recommendation is 'Manual Review' or the application exceeds the Loan Officer's approval authority, routes the application to the Manual Review Queue (SC-04) and notifies the Credit Committee (FR-NOT-02).
9	Credit Committee	Reviews the application and classification detail via the Manual Review Queue and records a final decision (Approve / Reject / Request More Information).
10	System	Logs the final decision, classification data, and rule version applied to the Audit Trail (FR-AUD-01) and updates the Applicant Record (FR-REC-01).
11	System	Notifies the Loan Officer of the final decision (FR-NOT-01).

PF-02: Business Rule Change and Approval

Step	Actor	Action / System Response
1	Risk Analyst	Accesses the Rule Configuration screen (SC-08) and creates or edits a business rule / threshold in draft status.
2	System	Validates the rule syntax and threshold values, and saves the change as a pending draft version.
3	System	Notifies the Risk Administrator that a rule change is pending approval (FR-NOT-04).
4	Risk Administrator	Reviews the proposed rule change, including a comparison with the current published version.
5	Risk Administrator	Approves or rejects the change. If rejected, the system notifies the Risk Analyst with the reason (FR-NOT-05).
6	System	On approval, publishes the new rule version to the production classification engine and archives the previous version with a timestamp and author (FR-ADM-04).
7	System	Logs the rule change event to the Audit Trail (FR-AUD-02).

PF-03: Manual Review Escalation

Step	Actor	Action / System Response
1	System	Identifies that an application's risk category and/or loan amount meets the escalation criteria defined in BR-07.
2	System	Places the application into the Manual Review Queue (SC-04) with a status of 'Pending Committee Review'.
3	System	Sends a notification to all active Credit Committee members (FR-NOT-02).

Step	Actor	Action / System Response
4	Credit Committee Member	Opens the application from the queue, reviews applicant data, risk classification, and contributing factors.
5	Credit Committee Member	Records a decision (Approve / Reject / Request More Information) with mandatory comments.
6	System	If 'Request More Information' is selected, returns the application to the originating Loan Officer with the committee's comments and notifies them (FR-NOT-03).
7	System	Updates application status and logs the decision to the Audit Trail.

PF-04: Report and Dashboard Generation

Step	Actor	Action / System Response
1	User (per role)	Navigates to the Dashboard (SC-09) or Reports screen (SC-10).
2	System	Displays default dashboard widgets / report list based on the user's role and access permissions.
3	User	Optionally applies filters (e.g., date range, branch, risk category, product type).
4	System	Refreshes visualizations and report data according to the applied filters.
5	User	Optionally exports a report to PDF or Excel format (FR-RPT-06).

5. Business Rules

The following business rules govern the behavior of the risk classification engine and decision workflow. Specific threshold values shown are illustrative placeholders to be finalized and formally approved by the Risk Management function prior to development; the rule structure and logic itself constitute the functional requirement.

ID	Business Rule Description
BR-01	An application shall not be submitted for risk classification unless all mandatory applicant fields (Section 6) are complete and pass validation.
BR-02	The system shall calculate a risk score for each applicant using a weighted combination of predefined risk factors, including but not limited to: income-to-obligation ratio, existing liabilities, repayment history (where available), employment stability, and loan-to-value or loan amount relative to income.
BR-03	The system shall classify the applicant into one of three risk categories based on the calculated risk score: Low Risk, Medium Risk, or High Risk, using thresholds configured and approved by the Risk Administrator.
BR-04	Where a customer has no prior relationship or repayment history with ABC Bank, the system shall rely on the available application and external credit bureau data and shall flag the classification as 'Limited History' for reviewer awareness.
BR-05	The system shall record the specific factors and their relative contribution that led to a given risk classification, to support the rationale shown to the user.
BR-06	The system shall generate a recommendation based on risk category as follows: Low Risk -> Approve; Medium Risk -> Manual Review; High Risk -> Reject, subject to override by an authorized Credit Committee decision.
BR-07	An application shall be automatically escalated to the Manual Review Queue if: (a) the risk category is Medium or High, or (b) the requested loan amount exceeds the Loan Officer's individual approval authority, whichever applies.
BR-08	A Loan Officer shall not be permitted to record a final 'Approve' decision on an application classified as High Risk; such applications must be decided by the Credit Committee.
BR-09	Business rule and threshold changes shall not take effect in the production classification engine until approved by an authorized Risk Administrator (segregation of duties from the Risk Analyst who proposed the change).
BR-10	Every risk classification result shall be retained together with the specific rule version applied, to ensure that historical decisions remain explainable even after rules are subsequently changed.
BR-11	If an applicant's financial data is amended before a final decision is recorded, the system shall automatically re-run the risk classification and supersede the prior result.
BR-12	The system shall not classify or process an application containing data that fails mandatory field validation (see Section 6); such applications shall remain in 'Draft' status.

6. Screen Specifications and Field Validations

This section defines the primary screens of the Financial Risk Classification System, their purpose, access roles, and key fields, followed by detailed field-level validation rules.

6.1 Screen Inventory

Screen ID	Screen Name	Accessible Roles	Purpose
SC-01	Login / Authentication	All Users	Authenticate user and route to role-based landing page
SC-02	Applicant Data Entry	Loan Officer	Capture applicant financial and demographic data for risk evaluation
SC-03	Risk Assessment Result	Loan Officer, Credit Committee, Compliance/Audit	Display risk category, contributing factors, and system recommendation
SC-04	Manual Review Queue	Credit Committee	List and action applications requiring committee review
SC-05	Decision Recording	Loan Officer, Credit Committee	Record the final decision on a loan application
SC-06	Applicant / Application Search	Loan Officer, Credit Committee, Compliance/Audit	Search and retrieve applicant records and application history
SC-07	User & Role Management	System Administrator	Create, edit, and deactivate user accounts and role assignments
SC-08	Rule Configuration	Risk Analyst, Risk Administrator	Define, review, and approve risk classification business rules and thresholds
SC-09	Dashboard	All Users (role-filtered)	Visual summary of application volumes, risk distribution, and trends
SC-10	Reports	All Users (role-filtered)	Generate, filter, and export management reports

SC-02: Applicant Data Entry - Key Fields

Field Name	Data Type	Mandatory	Validation Rule	Error ID
Applicant Full Name	Text	Yes	Alphabetic characters and spaces only; 2-100 characters	ERR-01
Date of Birth	Date	Yes	Valid date; applicant must be 18 years or older	ERR-02
National ID / Customer ID Number	Alphanumeric	Yes	Must match the bank's standard ID format; checked for duplicates	ERR-03

Field Name	Data Type	Mandatory	Validation Rule	Error ID
Employment Status	Dropdown	Yes	Must select one of the predefined values (Employed, Self-Employed, Unemployed, Retired)	ERR-04
Monthly Income	Numeric (currency)	Yes	Must be a positive number greater than zero; max 2 decimal places	ERR-05
Existing Monthly Obligations	Numeric (currency)	Yes	Must be zero or a positive number; max 2 decimal places	ERR-06
Requested Loan Amount	Numeric (currency)	Yes	Must be greater than zero and within the product's minimum/maximum limits	ERR-07
Loan Tenure (months)	Numeric (integer)	Yes	Must be within the product's allowable tenure range	ERR-08
Employer Name	Text	Conditional (if Employed/Self-Employed)	Required only when Employment Status = Employed or Self-Employed	ERR-09
Years at Current Employment	Numeric (integer)	Conditional (if Employed)	Must be zero or a positive whole number	ERR-10
Existing ABC Bank Customer?	Boolean (Yes/No)	Yes	Must be selected; determines historical data lookup (BR-04)	ERR-11
Supporting Documents	File Upload	Yes (at least 1)	Accepted formats: PDF, JPG, PNG; max 10MB per file	ERR-12

General Validation Behavior

- The system shall validate mandatory fields in real time as the user moves focus away from each field (inline validation), and again on form submission.
- The system shall prevent submission of the Applicant Data Entry form while any mandatory field validation is failing, and shall highlight all failing fields simultaneously upon a submit attempt.
- The system shall preserve entered data in the event of a validation failure so the user does not need to re-enter valid fields.
- The system shall display validation and error messages adjacent to the relevant field in plain, non-technical language.

7. Functional Requirements

7.1 Data Capture (FR-DC)

ID	Requirement	Priority
FR-DC-01	The system shall provide the Applicant Data Entry screen (SC-02) allowing a Loan Officer to capture applicant financial and demographic data required for risk classification.	Must Have
FR-DC-02	The system shall allow a Loan Officer to save an in-progress application as a draft prior to submission for classification.	Should Have
FR-DC-03	The system shall automatically pre-populate available fields (e.g., name, existing account details) for applicants who are existing ABC Bank customers, based on the Existing Customer indicator.	Should Have
FR-DC-04	The system shall allow the Loan Officer to attach supporting documents (e.g., proof of income, identification) to the application.	Must Have
FR-DC-05	The system shall timestamp and record the identity of the user who created and last modified each application record.	Must Have

7.2 Field Validation (FR-VAL)

ID	Requirement	Priority
FR-VAL-01	The system shall validate all mandatory fields defined in Section 6 before allowing an application to be submitted for risk classification.	Must Have
FR-VAL-02	The system shall enforce field-level format, type, and range validation as defined in the field validation table (Section 6).	Must Have
FR-VAL-03	The system shall display a specific, field-linked error message (Section 8) for each failed validation.	Must Have
FR-VAL-04	The system shall check for duplicate applicant/application records based on National ID / Customer ID and warn the Loan Officer before proceeding.	Should Have

7.3 Risk Classification Engine (FR-RCE)

ID	Requirement	Priority
FR-RCE-01	The system shall calculate a risk score for a submitted application by applying the predefined business rules and weighting defined in BR-02 to the validated applicant data.	Must Have
FR-RCE-02	The system shall classify the application into Low, Medium, or High Risk based on the calculated score and the thresholds defined in BR-03.	Must Have

ID	Requirement	Priority
FR-RCE-03	The system shall retrieve and incorporate available historical repayment and account data for existing ABC Bank customers into the risk calculation (BR-04).	Must Have
FR-RCE-04	The system shall record the individual contributing factors and their relative weight/impact for each risk classification result (BR-05).	Must Have
FR-RCE-05	The system shall generate a system recommendation (Approve / Manual Review / Reject) based on the risk category, per BR-06.	Must Have
FR-RCE-06	The system shall re-run the risk classification automatically if applicant financial data is amended prior to a final decision being recorded (BR-11).	Should Have
FR-RCE-07	The system shall apply the specific rule version that was active at the time of classification and record that version against the result (BR-10).	Must Have

7.4 Decision & Workflow (FR-DEC)

ID	Requirement	Priority
FR-DEC-01	The system shall display the Risk Assessment Result screen (SC-03) showing the risk category, score, contributing factors, and recommendation to the Loan Officer immediately after classification.	Must Have
FR-DEC-02	The system shall allow a Loan Officer to record a final decision directly for applications where the recommendation is within their approval authority (BR-08).	Must Have
FR-DEC-03	The system shall automatically route applications meeting escalation criteria (BR-07) to the Manual Review Queue (SC-04).	Must Have
FR-DEC-04	The system shall allow a Credit Committee member to record a decision of Approve, Reject, or Request More Information, with mandatory comments, from the Manual Review Queue.	Must Have
FR-DEC-05	The system shall prevent a Loan Officer from recording an 'Approve' decision on a High Risk application (BR-08), presenting the decision options as read-only with an explanatory message in that case.	Must Have
FR-DEC-06	The system shall return applications marked 'Request More Information' to the originating Loan Officer along with the committee's comments.	Must Have
FR-DEC-07	The system shall update the application status (e.g., Draft, Submitted, Pending Review, Approved, Rejected, Info Requested) at each step of the workflow and display the current status to relevant users.	Must Have

7.5 Applicant Records Management (FR-REC)

ID	Requirement	Priority
FR-REC-01	The system shall maintain a complete record of each application, including submitted data, classification results, decisions, and status history.	Must Have

ID	Requirement	Priority
FR-REC-02	The system shall provide a search function (SC-06) allowing authorized users to retrieve applicant and application records by criteria including applicant name, ID number, application ID, status, or date range.	Must Have
FR-REC-03	The system shall display the full history of a given applicant's prior applications and outcomes, where the user has appropriate access.	Should Have
FR-REC-04	The system shall retain application and classification records for the period defined by ABC Bank's data retention policy.	Must Have

7.6 Reporting and Dashboards (FR-RPT)

ID	Requirement	Priority
FR-RPT-01	The system shall provide a Dashboard (SC-09) summarizing loan application volumes, risk category distribution, and decision outcomes for a selected period.	Must Have
FR-RPT-02	The system shall allow dashboard data to be filtered by date range, branch, product type, and risk category.	Should Have
FR-RPT-03	The system shall provide a standard set of reports (Section 9) accessible via the Reports screen (SC-10).	Must Have
FR-RPT-04	The system shall restrict dashboard and report data visibility according to the user's role and, where applicable, branch assignment.	Must Have
FR-RPT-05	The system shall allow authorized users to export reports in PDF and Excel formats.	Should Have
FR-RPT-06	The system shall refresh dashboard data at a defined interval or on user-triggered refresh, with the last-updated time displayed.	Could Have

7.7 Administration & Rule Configuration (FR-ADM)

ID	Requirement	Priority
FR-ADM-01	The system shall allow a System Administrator to create, edit, deactivate, and reassign roles for user accounts (SC-07).	Must Have
FR-ADM-02	The system shall allow a Risk Analyst to create or edit risk classification business rules and thresholds in a draft/pending state (SC-08).	Must Have
FR-ADM-03	The system shall require Risk Administrator approval before any rule change is published to the production classification engine (BR-09).	Must Have
FR-ADM-04	The system shall retain a version history of all published rule sets, including author, approver, timestamp, and change description.	Must Have
FR-ADM-05	The system shall allow a Risk Administrator to view a comparison between a proposed rule version and the currently published version prior to approval.	Should Have

7.8 Notifications (FR-NOT)

ID	Requirement	Priority
FR-NOT-01	The system shall notify the originating Loan Officer when a final decision has been recorded on their submitted application.	Must Have
FR-NOT-02	The system shall notify active Credit Committee members when an application is routed to the Manual Review Queue.	Must Have
FR-NOT-03	The system shall notify the originating Loan Officer when an application is returned with a 'Request More Information' status.	Must Have
FR-NOT-04	The system shall notify the Risk Administrator when a business rule change is submitted for approval.	Must Have
FR-NOT-05	The system shall notify the Risk Analyst when their proposed rule change is approved or rejected, including rejection reason where applicable.	Should Have

7.9 Audit Trail (FR-AUD)

ID	Requirement	Priority
FR-AUD-01	The system shall log every risk classification event, including input data snapshot, rule version applied, result, and timestamp.	Must Have
FR-AUD-02	The system shall log every business rule change event, including author, approver, previous and new values, and timestamp.	Must Have
FR-AUD-03	The system shall log every decision recorded against an application, including the user, decision, comments, and timestamp.	Must Have
FR-AUD-04	The system shall provide read-only audit log access to Compliance/Audit users, with search and filter capability.	Must Have
FR-AUD-05	The system shall ensure audit log entries cannot be edited or deleted by any user role.	Must Have

8. Error Messages

The following table defines the standard system-generated error and validation messages referenced throughout this document.

Error ID	Trigger Condition	Message Text (Displayed to User)	Type
ERR-01	Applicant Full Name is blank or contains invalid characters	Please enter a valid full name using letters only.	Validation
ERR-02	Date of Birth is blank, invalid, or applicant is under 18	Please enter a valid date of birth. Applicant must be at least 18 years old.	Validation
ERR-03	National ID / Customer ID is blank, invalid format, or duplicate	Please enter a valid ID number. This ID already exists in the system.	Validation
ERR-04	Employment Status not selected	Please select an employment status.	Validation
ERR-05	Monthly Income is blank, zero, or negative	Please enter a monthly income greater than zero.	Validation
ERR-06	Existing Monthly Obligations is negative	Existing obligations cannot be a negative value.	Validation
ERR-07	Requested Loan Amount is outside product limits	Loan amount must be between [minimum] and [maximum] for the selected product.	Validation
ERR-08	Loan Tenure outside allowable range	Loan tenure must be between [min] and [max] months for the selected product.	Validation
ERR-09	Employer Name blank when Employment Status requires it	Please enter an employer name.	Validation
ERR-10	Years at Current Employment blank or negative	Please enter a valid number of years (zero or greater).	Validation
ERR-11	Existing Customer indicator not selected	Please indicate whether the applicant is an existing ABC Bank customer.	Validation
ERR-12	Supporting document missing, invalid format, or exceeds size limit	Please attach at least one supporting document (PDF, JPG, or PNG, max 10MB).	Validation
ERR-13	Risk classification engine unable to process due to a system/data error	The risk assessment could not be completed. Please try again or contact support.	System
ERR-14	User attempts an action outside their role's permissions	You do not have permission to perform this action.	System / Security
ERR-15	Session has expired due to inactivity	Your session has expired. Please log in again.	System / Security
ERR-16	Attempt to approve a High Risk application by a Loan Officer	High Risk applications require Credit Committee approval and cannot be approved at this level.	Business Rule

Error ID	Trigger Condition	Message Text (Displayed to User)	Type
ERR-17	Attempt to publish a rule change without Risk Administrator approval	This rule change must be approved by a Risk Administrator before it can be published.	Business Rule

9. Reports and Dashboards

Report ID	Report Name	Purpose	Key Data Elements	Typical Audience
RPT-01	Loan Application Volume Summary	Track total applications received, processed, and pending over a period	Date, Branch, Product, Application Count, Status	Branch Manager, Credit / Lending Management
RPT-02	Risk Distribution Report	Show the distribution of applications across Low, Medium, and High Risk categories	Risk Category, Count, Percentage, Period, Branch	Risk Management, Credit Committee
RPT-03	Decision Outcome Report	Compare system recommendations against final decisions made	Recommendation, Final Decision, Match/Override Rate, Decision Maker	Risk Management, Compliance
RPT-04	Manual Review Turnaround Report	Measure time taken for Credit Committee review and decision	Application ID, Date Escalated, Date Decided, Turnaround Time	Credit / Lending Management
RPT-05	Business Rule Change History Report	List all rule changes, approvals, and effective dates	Rule ID, Change Description, Author, Approver, Effective Date	Risk Management, Compliance / Audit
RPT-06	Audit Log Export	Provide a detailed, exportable log of classification and decision events for a given period or application	Event Type, User, Timestamp, Before/After Values	Compliance / Audit, Internal Audit

9.1 Dashboard Widgets (SC-09)

- Application volume trend (line/bar chart) by day, week, or month.
- Risk category distribution (pie/donut chart) for the selected period.
- Recommendation vs. final decision override rate (indicator/summary tile).
- Applications pending Manual Review (count with quick-access link to the queue).
- Top contributing risk factors across recent classifications (summary list).

10. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Performance	The system shall return a risk classification result within 5 seconds of application submission under normal operating load.
NFR-02	Performance	Dashboard and report screens shall load within 8 seconds for standard date ranges (up to 12 months) under normal operating load.
NFR-03	Availability	The system shall be available 99.5% of the time during ABC Bank's standard business operating hours, excluding scheduled maintenance.
NFR-04	Security	All user access shall require authentication, and functionality/data visibility shall be enforced according to role-based access control (RBAC).
NFR-05	Security	Sensitive applicant financial data shall be encrypted in transit (TLS 1.2 or higher) and at rest.
NFR-06	Auditability	All classification decisions, rule changes, and user decisions shall be logged in a manner that is tamper-evident and cannot be altered or deleted.
NFR-07	Usability	Risk classification results and contributing factors shall be presented in plain, non-technical language understandable to a Loan Officer without risk-modeling expertise.
NFR-08	Accessibility	User interface screens shall conform to the bank's internal accessibility standards for internal-use business applications.
NFR-09	Scalability	The system shall support the current and projected loan application volumes (as defined during capacity planning) without performance degradation.
NFR-10	Reliability	The risk classification engine shall produce identical classification output for identical input data and rule version (deterministic behavior).
NFR-11	Data Retention	Application, classification, and audit records shall be retained per ABC Bank's data retention policy and shall be recoverable within defined backup/restore objectives.
NFR-12	Compatibility	The system's user interface shall function correctly on the standard desktop browser(s) approved for internal use at ABC Bank.
NFR-13	Maintainability	Business rules and thresholds shall be configurable by authorized users through the Rule Configuration screen without requiring source code changes.

11. Assumptions and Constraints

11.1 Assumptions

- All business rule threshold values referenced in Section 5 will be finalized and formally approved by Risk Management prior to development sign-off.
- Historical repayment data for existing customers is accessible from existing internal systems in a consumable format.
- Users will access the system using bank-managed devices already covered by existing authentication and security infrastructure.
- Loan product limits (minimum/maximum amount and tenure) referenced in field validations are maintained and available as reference data.

11.2 Constraints

- The system must integrate with existing core banking and credit bureau data sources rather than duplicate this data independently.
- Any change to escalation thresholds or approval authority levels must go through the change control process defined for business rules (BR-09).
- The solution must comply with ABC Bank's existing data security and privacy policies referenced in Section 2.3.

12. Acceptance Criteria

Acceptance criteria below are expressed in Given/When/Then format for key functional requirement groups and will be elaborated further into detailed test cases during the test planning phase.

AC-01: Applicant Data Validation (ref. FR-VAL-01, FR-VAL-02)

- Given a Loan Officer is completing the Applicant Data Entry screen, when a mandatory field is left blank and the form is submitted, then the system shall block submission and display the corresponding error message (Section 8) next to the field.
- Given all mandatory fields are completed with valid data, when the form is submitted, then the system shall accept the application and proceed to risk classification.

AC-02: Risk Classification Result (ref. FR-RCE-01 to FR-RCE-05)

- Given a validly submitted application, when the system completes risk classification, then the application shall be assigned exactly one risk category (Low, Medium, or High) and a corresponding recommendation, consistent with BR-06.
- Given a completed risk classification, when the Loan Officer views the Risk Assessment Result screen, then the system shall display the risk category, score, and the specific contributing factors used to determine the result.

AC-03: Decision Authority Enforcement (ref. FR-DEC-02, FR-DEC-05, BR-08)

- Given an application classified as High Risk, when a Loan Officer views the Decision Recording screen, then the 'Approve' option shall not be available to the Loan Officer, and the application shall be marked for Credit Committee decision.
- Given an application classified as Low Risk within the Loan Officer's approval authority, when the Loan Officer records an 'Approve' decision, then the system shall accept the decision and update the application status accordingly.

AC-04: Manual Review Escalation (ref. FR-DEC-03, BR-07)

- Given an application meets the escalation criteria in BR-07, when risk classification completes, then the application shall automatically appear in the Manual Review Queue and Credit Committee members shall receive a notification.

AC-05: Business Rule Change Control (ref. FR-ADM-02, FR-ADM-03, BR-09)

- Given a Risk Analyst creates or edits a business rule, when the change is saved, then it shall remain in a pending/draft state and shall not affect production classification results until approved.

- Given a Risk Administrator approves a pending rule change, when the approval is confirmed, then the new rule version shall be published and applied to all subsequent classifications, and the previous version shall be archived with a version record.

AC-06: Audit Trail Integrity (ref. FR-AUD-01 to FR-AUD-05)

- Given any classification, decision, or rule change event occurs, when the event completes, then a corresponding audit log entry shall be created capturing the user, timestamp, and relevant before/after data.
- Given an audit log entry exists, when any user (including administrators) attempts to edit or delete it through the application, then the system shall prevent the action.

AC-07: Reporting Access Control (ref. FR-RPT-04)

- Given a user without Compliance/Audit or management role, when they access the Reports screen, then only reports and data permitted for their role and branch assignment shall be visible.

13. Requirements Traceability Summary

The table below provides a high-level mapping from the BRD business objectives to the corresponding functional requirement groups defined in this document, to support end-to-end traceability.

BRD Business Objective	Related Functional Requirement Group(s)
Reduce loan default rate through improved risk assessment accuracy	FR-RCE (Risk Classification Engine), Business Rules Section 5
Reduce average loan evaluation turnaround time	FR-DC (Data Capture), FR-VAL (Validation), FR-RCE (Classification), PF-01
Standardize risk evaluation criteria across branches	FR-RCE, FR-ADM (Rule Configuration), Business Rules Section 5
Provide explainable risk classification to support decisions	FR-RCE-04, FR-DEC-01 (Risk Assessment Result screen)
Maintain auditable, transparent evaluation process	FR-AUD (Audit Trail), FR-ADM-03/04 (Rule Change Control)
Provide management visibility into risk and application trends	FR-RPT (Reporting & Dashboards), Section 9

14. Glossary

Term	Definition
FRD	Functional Requirements Document - this document.
Risk Score	A numeric value calculated by the risk classification engine representing the applicant's assessed level of credit risk.
Risk Category	The classification bucket (Low, Medium, High) assigned to an applicant based on the risk score and configured thresholds.
Recommendation	The system-generated suggested outcome (Approve, Manual Review, Reject) based on the risk category.
Manual Review Queue	A worklist of applications requiring Credit Committee review before a final decision is recorded.
RBAC	Role-Based Access Control - restricting system functionality and data based on the user's assigned role.
Draft Rule	A business rule or threshold change that has been created or edited but not yet approved for production use.
Audit Trail	A chronological, immutable log of system events including classifications, decisions, and configuration changes.